

Requirement Understanding Document

Community Library Membership & Book Lending System

1. Objectives

1. Manage library members
2. Maintain books and copies
3. Automate borrowing and returns
4. Calculate fines
5. Generate reports
6. Implement authentication and authorization

2. Functional Requirements

2.1 Authentication Module

Roles:

- Admin/Librarian
- User/Member

Admin can:

- Add members
- Deactivate members
- Add books and copies
- View reports
- Manage fines

Users can:

- View books
- Borrow books
- Return books
- View and pay fines

2.2 Member Management

- Add member
- View all members
- Search member
- Update status

- Deactivate member

Membership Types:

- Basic
- Student
- Premium

2.3 Membership Rules

Basic:

- Max Books: 2
- Max Days: 7

Student:

- Max Books: 3
- Max Days: 10

Premium:

- Max Books: 5
- Max Days: 15

2.4 Book Management

- Add books
- Add copies
- View available books
- Search books
- Mark damaged/lost books

2.5 Borrowing Rules

Borrowing blocked if:

- Member inactive
- Borrow limit exceeded
- Pending fines above ₹500
- No available copies
- Duplicate borrowing

2.6 Return Rules

- Return date updated
- Late fine calculated
- Damage fine calculated
- Book copy updated
- Fine record created

2.7 Fine Management

- View pending fines
- Pay fine
- View fine history

2.8 Reports

- Available books by category
- Overdue books
- Most borrowed books
- Pending fines
- Member borrowing summary

3. Database Design

Main Entities:

- User
- Member
- MembershipType
- Category
- Book
- BookCopy
- Borrowing
- Fine
- FinePayment

Relationships:

- MembershipType → Members
- Category → Books
- Book → BookCopies
- Member → Borrowings
- Borrowing → Fine

4. Business Rules

- Only active members can borrow
- Fine above ₹500 blocks borrowing
- Fully damaged books marked lost
- Borrowing transactions use rollback

5. PostgreSQL Functions and Procedures

Functions:

- get_available_books_report()
- get_overdue_books_report()
- get_member_borrowing_summary()
- get_members_with_pending_fines()
- get_most_borrowed_books()

Procedure/Function:

- process_book_return()

6. Transaction Management

Transactions are implemented during:

- Borrow book
- Return book

Rollback occurs if any step fails.

7. Exception Handling

Custom validations:

- Invalid member
- Invalid book
- No available copies
- Fine limit exceeded
- Borrowing limit exceeded
- Duplicate borrowing

8. Conclusion

The project demonstrates real-world implementation of EF Core, PostgreSQL, layered architecture, transaction handling, reporting, and business rule enforcement in a library management system.